

From System of Record to System of Agency: Designing the Experience

Executive Summary

The next generation of enterprise software is not a destination users visit; it is an ambient force that works on their behalf. By moving toward a **Headless ERP**, we are shifting the human role from "Data Entry Operator" to "Strategic Orchestrator." This document defines the new design paradigms required to manage people, money, and agents in an ecosystem where the machine is always active and trust is the primary currency.

I. Core Design Paradigms

1. Intent-Based Orchestration

We are replacing "Point-and-Click" workflows with "Outcome-and-Verify" logic.

- **The Concept:** Users define high-level business goals (e.g., "Optimize Q3 cash flow while maintaining 98% vendor satisfaction").
- **Design Solution: Constrained Natural Language templates** that help users frame intent with clear parameters.
- **Shadow Execution:** Before a multi-step agentic task begins, the system provides a preview of the logic, allowing the user to sign off on the *intent* before the *action*.

2. Intervention-Driven UI (The "Quiet UI")

In an ambient system, a silent interface is a sign of a high-functioning system.

- **The Concept:** The UI only demands attention for high-stakes decisions or anomalies.
- **Design Solution: Situational Briefings.** Instead of menus, the entry point is a timeline of agent actions. We use a "Traffic Light" status:
 - **Green:** Automated & Successful.
 - **Yellow:** Needs Review / Divergence detected.
 - **Red:** Blocked / Conflict requiring human intuition.

3. Liquid Presence & Atomic Components

The ERP has no "home base." It lives where the work happens.

- **The Concept:** UI must be atomic, capable of living in Slack, Teams, Mobile, or within 3rd party tools.
- **Design Solution: Micro-UIs.** Self-contained widgets that carry full data integrity and action capability into external workflows, allowing for "in-stream" approvals without context switching.

II. The Logic of Trust & Liability

4. Calibrated Friction & The 100% Rule

To solve the "Ghost in the Machine" problem, we implement intentional friction.

- **High-Stakes Sensitivity:** For actions involving payroll, large transfers, or personnel changes, the system requires **100% Confidence**. Any score below 100% triggers a mandatory Human-in-the-Loop (HITL) "Witness" step.
- **The Impact Slider:** Administrators can define "Agency Thresholds." (e.g., "Automate all spend < \$500; I must witness all spend > \$10k regardless of confidence.")

5. Semantic Traceability (Reasoning Trees)

Trust is built through observability, not just results.

- **The Concept:** Every automated outcome must be auditable in plain language.
- **Design Solution: Reasoning Trees.** A visual map showing:
[Data Input] → [Applied Policy] → [Calculation] → [Proposed Action]
This allows a human to audit the "Why" behind an agent's decision at a glance.

6. The Universal Kill Switch

In a multi-agent, headless system, the ability to "Stop All Stations" is a safety requirement.

- **Design Solution:** A persistent, high-contrast emergency control accessible across all surfaces (Mobile, Web, API headers). Activation instantly pauses all active "Write" permissions for agents while maintaining "Read" status for diagnostic audit.

III. The Relational Experience

7. Relational Onboarding & Mentor Mode

Onboarding is no longer a checklist; it is a developing partnership between human and machine.

- **Mentor Mode:** The AI proactively surfaces "Capability Briefs" based on user behavior.
- **Mirroring Mode:** During initial setup, the agent operates in a "ghost" state—showing what it *would* do in a parallel stream—allowing the user to gain confidence before granting autonomy.

8. Adaptive Contextual Presence

The AI persona must scale its presence to match user proficiency.

- **The Concept:** The system should know when a user is struggling versus when they are in a "flow state."
- **Design Solution: Sensitivity Dials.** Users can tune the AI's "Proactivity Level."
 - *High Sensitivity:* AI offers suggestions and logic explanations frequently (Early stage/New features).
 - *Low Sensitivity:* AI only interrupts for critical exceptions (Power user/Routine operations).

IV. The Agentic Marketplace

9. 1st and 3rd Party Governance

As we accept external agents, the ERP becomes an Operating System for Agency.

- **The Concept:** Clear distinction between internal (verified) and external (guest) agents.
- **Design Solution: The Credentials Card.** Every action is stamped with a "Source of Agency." 3rd party agents carry a "Passport" showing their permissions, reliability scores, and mission parameters.

10. Multi-Agent Conflict Resolution

When two agents have conflicting goals (e.g., Savings Agent vs. Growth Agent), the system reaches a stalemate.

- **Design Solution: The Mediation UI.** The system presents the conflict to the Admin/HITL. It summarizes the two opposing arguments and the projected outcomes of each, allowing the human to act as the "Tie-Breaker."

Conclusion

Building a Headless ERP is not a technical challenge; it is a trust challenge. By focusing on **Calibrated Friction**, **Semantic Traceability**, and **Relational Intelligence**, we ensure that while the machine is ambient, the human remains the ultimate authority.